

**Contents**

# 1 Orbitals And Their Symmetry

We use the following order of orbitals here (assuming degeneracy of LUMOs and HOMOs):

$$\begin{array}{cc} b_{1u} & \text{---} & \text{---} & a_u \\ b_{2g} & \text{---} & \text{---} & b_{3g} \\ \\ b_{1u} & \text{---} & \text{---} & a_u \\ b_{2g} & \text{---} & \text{---} & b_{3g} \end{array}$$

In the following, we will leave out the explicit labeling by orbital symmetry. Each monomer will simply be written as:

$$\begin{array}{cc} \text{---} & \text{---} \\ \text{---} & \text{---} \end{array}$$

where the character of the orbital is defined by its position.

Dimer configurations will be written as their monomer occupations, by writing one monomer to the left and one to the right. The orbital order within the group of monomer orbitals follows the above mentioned definition.

The transformation of monomer orbitals into dimer orbitals is given by knowing the negative and positive linear combinations of monomer orbitals into dimer orbitals. Transforming this known equation system leads to:

$$a_u^l = +b_{1g} + a_u \tag{1a}$$

$$b_{1u}^l = +a_g + b_{1u} \tag{1b}$$

$$b_{2g}^l = +b_{3u} + b_{2g} \tag{1c}$$

$$b_{3g}^l = +b_{2u} + b_{3g} \tag{1d}$$

$$a_u^r = -b_{1g} + a_u \tag{1e}$$

$$b_{1u}^r = -a_g + b_{1u} \tag{1f}$$

$$b_{2g}^r = -b_{3u} + b_{2g} \tag{1g}$$

$$b_{3g}^r = -b_{2u} + b_{3g} \tag{1h}$$

## 2 Monomer States and Configurations

$$i^3b_{2u} = + \begin{pmatrix} \uparrow & - \\ \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow \\ \uparrow & \downarrow\uparrow \end{pmatrix} = |a2a0| - |0a2a|$$

$$e^3b_{2u} = + \begin{pmatrix} \uparrow & - \\ \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow \\ \uparrow & \downarrow\uparrow \end{pmatrix} = |a2a0| + |0a2a|$$

$$e^3b_{3u} = + \begin{pmatrix} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow \\ \downarrow\uparrow & \uparrow \end{pmatrix} = |aa20| - |02aa|$$

$$i^3b_{3u} = + \begin{pmatrix} \uparrow & - \\ \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow \\ \downarrow\uparrow & \uparrow \end{pmatrix} = |aa20| + |02aa|$$

### 3 Linear Combinations: 16 Monomer- / 8 Dimer-States

identical  $i^3 b_{2u} * i^3 b_{2u} =$  identical 1.3 \* 1.3

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}$$

written in CI Vectors:

$$+1 \cdot |a2a0| \cdot |a2a0| - 1 \cdot |0a2a| \cdot |a2a0| - 1 \cdot |a2a0| \cdot |0a2a| + 1 \cdot |0a2a| \cdot |0a2a|$$

forms the following possible dimer combinations:

$$\begin{aligned} & +1 \cdot |a2a0a2a0| + 1 \cdot |0a2a0a2a| - \frac{1}{8} \cdot |0a2aa2a0| + \frac{1}{8} \cdot |0a20a2aa| + \frac{1}{8} \cdot |0aaaa220| - \frac{1}{8} \cdot |0aa0a22a| \\ & + \frac{1}{8} \cdot |022aaaa0| - \frac{1}{8} \cdot |0220aaaa| - \frac{1}{8} \cdot |02aaaa20| + \frac{1}{8} \cdot |02a0aa2a| + \frac{1}{8} \cdot |aa2a02a0| - \frac{1}{8} \cdot |aa2002aa| \\ & - \frac{1}{8} \cdot |aaaa0220| + \frac{1}{8} \cdot |aaa0022a| - \frac{1}{8} \cdot |a22a0aa0| + \frac{1}{8} \cdot |a2200aaa| + \frac{1}{8} \cdot |a2aa0a20| - \frac{1}{8} \cdot |a2a00a2a| \end{aligned}$$

or, expressed in terms of symmetry:

$$\begin{aligned} = & -4 \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right| - 2 \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + 2 \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + 2 \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\ & + 2 \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + 2 \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 2 \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + 2 \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\ & - 2 \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - 4 \left| \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g} \right| \end{aligned}$$

plus combination:  $i^3 b_{2u} * e^3 b_{2u} + e^3 b_{2u} * i^3 b_{2u} =$  plus combination: 1.3 \* 2.3 + 2.3 \* 1.3

$$\begin{aligned} & + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} \\ & + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} \end{aligned}$$

written in CI Vectors:

$$+1 \cdot |a2a0| \cdot |a2a0| - 1 \cdot |0a2a| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |0a2a| - 1 \cdot |0a2a| \cdot |0a2a|$$

$$+1 \cdot |a2a0| \cdot |a2a0| + 1 \cdot |0a2a| \cdot |a2a0| - 1 \cdot |a2a0| \cdot |0a2a| - 1 \cdot |0a2a| \cdot |0a2a|$$

forms the following possible dimer combinations:

$$+2 \cdot |a2a0a2a0| - 2 \cdot |0a2a0a2a|$$

or, expressed in terms of symmetry:

$$= -8 \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right| + 8 \left| \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g} \right|$$


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minus combination:  $i^3 b_{2u} * e^3 b_{2u} - e^3 b_{2u} * i^3 b_{2u} =$  minus combination: 1.3 \* 2.3 - 2.3 \* 1.3

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} \\ - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

written in CI Vectors:

$$+1 \cdot |a2a0| \cdot |a2a0| - 1 \cdot |0a2a| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |0a2a| - 1 \cdot |0a2a| \cdot |0a2a| \\ -1 \cdot |a2a0| \cdot |a2a0| - 1 \cdot |0a2a| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |0a2a| + 1 \cdot |0a2a| \cdot |0a2a|$$

forms the following possible dimer combinations:

or, expressed in terms of symmetry:

$$= -8 \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right| + 8 \left| \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g} \right|$$


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plus combination:  $i^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * i^3 b_{2u} =$  plus combination: 1.3 \* 1.2 + 1.2 \* 1.3

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} \\ + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\uparrow & \uparrow & \downarrow\uparrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\uparrow & \uparrow & \downarrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\uparrow & \uparrow & \uparrow & \downarrow\uparrow \end{pmatrix}$$

written in CI Vectors:

$$+1 \cdot |a2a0| \cdot |aa20| - 1 \cdot |0a2a| \cdot |aa20| - 1 \cdot |a2a0| \cdot |02aa| + 1 \cdot |0a2a| \cdot |02aa| \\ +1 \cdot |aa20| \cdot |a2a0| - 1 \cdot |02aa| \cdot |a2a0| - 1 \cdot |aa20| \cdot |0a2a| + 1 \cdot |02aa| \cdot |0a2a|$$

forms the following possible dimer combinations:

$$+\frac{1}{2} \cdot |a2a0aa20| + \frac{1}{2} \cdot |a220aaa0| + \frac{1}{2} \cdot |aaa0a220| + \frac{1}{2} \cdot |aa20a2a0| + \frac{1}{2} \cdot |0a2a02aa| + \frac{1}{2} \cdot |0aaa022a| \\ + \frac{1}{2} \cdot |022a0aaa| + \frac{1}{2} \cdot |02aa0a2a|$$

or, expressed in terms of symmetry:

$$\begin{aligned}
&= -4 \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \right| + 4 \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| - 4 \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| + 4 \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right| \\
&\quad - 4 \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} \right| + 4 \left| \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} \right| - 4 \left| \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} \right| + 4 \left| \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \right|
\end{aligned}$$


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minus combination:  $i^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * i^3 b_{2u} =$  minus combination:  $1.3 * 1.2 - 1.2 * 1.3$

$$\begin{aligned}
&+ \left( \begin{array}{cc} \uparrow & - \\ \downarrow\downarrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\downarrow \end{array} \right) - \left( \begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow\downarrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\downarrow \end{array} \right) - \left( \begin{array}{cc} \uparrow & - \\ \downarrow\downarrow & \uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \downarrow\downarrow & \uparrow \end{array} \right) + \left( \begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow\downarrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \downarrow\downarrow & \uparrow \end{array} \right) \\
&- \left( \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\downarrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \downarrow\downarrow & \uparrow \end{array} \right) + \left( \begin{array}{cc} - & \uparrow \\ \downarrow\downarrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \downarrow\downarrow & \uparrow \end{array} \right) + \left( \begin{array}{cc} \uparrow & - \\ \uparrow & \downarrow\downarrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \downarrow\downarrow & \uparrow \end{array} \right) - \left( \begin{array}{cc} - & \uparrow \\ \downarrow\downarrow & \uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow & \downarrow\downarrow \end{array} \right)
\end{aligned}$$

written in CI Vectors:

$$\begin{aligned}
&+1 \cdot |a2a0| \cdot |aa20| - 1 \cdot |0a2a| \cdot |aa20| - 1 \cdot |a2a0| \cdot |02aa| + 1 \cdot |0a2a| \cdot |02aa| \\
&-1 \cdot |aa20| \cdot |a2a0| + 1 \cdot |02aa| \cdot |a2a0| + 1 \cdot |aa20| \cdot |0a2a| - 1 \cdot |02aa| \cdot |0a2a|
\end{aligned}$$

forms the following possible dimer combinations:

$$\begin{aligned}
&+ \frac{1}{2} \cdot |0a2aaa20| - \frac{1}{2} \cdot |0a20aa2a| + \frac{1}{2} \cdot |aa2a0a20| - \frac{1}{2} \cdot |aa200a2a| + \frac{1}{2} \cdot |a2a002aa| - \frac{1}{2} \cdot |a2aa02a0| \\
&+ \frac{1}{2} \cdot |02a0a2aa| - \frac{1}{2} \cdot |02aaa2a0|
\end{aligned}$$

or, expressed in terms of symmetry:

$$\begin{aligned}
&= -4 \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \right| + 4 \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| - 4 \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| + 4 \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right| \\
&\quad - 4 \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} \right| + 4 \left| \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} \right| - 4 \left| \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} \right| + 4 \left| \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \right|
\end{aligned}$$


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plus combination:  $i^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * i^3 b_{2u} =$  plus combination:  $1.3 * 2.2 + 2.2 * 1.3$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix}$$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI Vectors:

$$+1 \cdot |a2a0| \cdot |aa20| - 1 \cdot |0a2a| \cdot |aa20| + 1 \cdot |a2a0| \cdot |02aa| - 1 \cdot |0a2a| \cdot |02aa|$$

$$+1 \cdot |aa20| \cdot |a2a0| + 1 \cdot |02aa| \cdot |a2a0| - 1 \cdot |aa20| \cdot |0a2a| - 1 \cdot |02aa| \cdot |0a2a|$$

forms the following possible dimer combinations:

$$+\frac{1}{2} \cdot |a2a0aa20| + \frac{1}{2} \cdot |a220aaa0| + \frac{1}{2} \cdot |aaa0a220| + \frac{1}{2} \cdot |aa20a2a0| - \frac{1}{2} \cdot |0a2a02aa| - \frac{1}{2} \cdot |0aaa022a|$$

$$-\frac{1}{2} \cdot |022a0aaa| - \frac{1}{2} \cdot |02aa0a2a|$$

or, expressed in terms of symmetry:

$$= -4 \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \right| + 4 \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| - 4 \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| + 4 \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right|$$

$$+ 4 \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} \right| - 4 \left| \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} \right| + 4 \left| \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} \right| - 4 \left| \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \right|$$

minus combination:  $i^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * i^3 b_{2u} =$  minus combination:  $1.3 * 2.2 - 2.2 * 1.3$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix}$$

$$- \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI Vectors:

$$+1 \cdot |a2a0| \cdot |aa20| - 1 \cdot |0a2a| \cdot |aa20| + 1 \cdot |a2a0| \cdot |02aa| - 1 \cdot |0a2a| \cdot |02aa|$$

$$-1 \cdot |aa20| \cdot |a2a0| - 1 \cdot |02aa| \cdot |a2a0| + 1 \cdot |aa20| \cdot |0a2a| + 1 \cdot |02aa| \cdot |0a2a|$$

forms the following possible dimer combinations:

$$+\frac{1}{2} \cdot |0a2aaa20| - \frac{1}{2} \cdot |0a20aa2a| + \frac{1}{2} \cdot |aa2a0a20| - \frac{1}{2} \cdot |aa200a2a| - \frac{1}{2} \cdot |a2a002aa| + \frac{1}{2} \cdot |a2aa02a0|$$

$$-\frac{1}{2} \cdot |02a0a2aa| + \frac{1}{2} \cdot |02aaa2a0|$$

or, expressed in terms of symmetry:

$$\begin{aligned}
&= -4 \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \right| + 4 \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| - 4 \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| + 4 \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right| \\
&\quad + 4 \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} \right| - 4 \left| \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} \right| + 4 \left| \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} \right| - 4 \left| \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \right|
\end{aligned}$$


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identical  $e^3 b_{2u} * e^3 b_{2u}$  = identical 2.3 \* 2.3

$$+ \left( \begin{array}{cc} \uparrow & - \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow\uparrow & \uparrow \end{array} \right) + \left( \begin{array}{cc} - & \uparrow \\ \uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} \uparrow & - \\ \uparrow\uparrow & \uparrow \end{array} \right) + \left( \begin{array}{cc} \uparrow & - \\ \uparrow\uparrow & \uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow & \uparrow\uparrow \end{array} \right) + \left( \begin{array}{cc} - & \uparrow \\ \uparrow & \uparrow\uparrow \end{array} \quad \begin{array}{cc} - & \uparrow \\ \uparrow & \uparrow\uparrow \end{array} \right)$$

written in CI Vectors:

$$+1 \cdot |a2a0| \cdot |a2a0| + 1 \cdot |0a2a| \cdot |a2a0| + 1 \cdot |a2a0| \cdot |0a2a| + 1 \cdot |0a2a| \cdot |0a2a|$$

forms the following possible dimer combinations:

$$\begin{aligned}
&+1 \cdot |a2a0a2a0| + 1 \cdot |0a2a0a2a| + \frac{1}{8} \cdot |0a2aa2a0| - \frac{1}{8} \cdot |0a20a2aa| - \frac{1}{8} \cdot |0aaaa220| + \frac{1}{8} \cdot |0aa0a22a| \\
&- \frac{1}{8} \cdot |022aaaa0| + \frac{1}{8} \cdot |0220aaaa| + \frac{1}{8} \cdot |02aaaa20| - \frac{1}{8} \cdot |02a0aa2a| - \frac{1}{8} \cdot |aa2a02a0| + \frac{1}{8} \cdot |aa2002aa| \\
&+ \frac{1}{8} \cdot |aaaa0220| - \frac{1}{8} \cdot |aaa0022a| + \frac{1}{8} \cdot |a22a0aa0| - \frac{1}{8} \cdot |a2200aaa| - \frac{1}{8} \cdot |a2aa0a20| + \frac{1}{8} \cdot |a2a00a2a|
\end{aligned}$$

or, expressed in terms of symmetry:

$$\begin{aligned}
&= -4 \left| \underbrace{a_g b_{2u} b_{1u} b_{3g}}_{a_g} \right| + 2 \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - 2 \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - 2 \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\
&\quad - 2 \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - 2 \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - 2 \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - 2 \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\
&\quad + 2 \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - 4 \left| \underbrace{b_{3u} b_{1g} b_{2g} a_u}_{a_g} \right|
\end{aligned}$$


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plus combination:  $e^3 b_{2u} * e^3 b_{3u} + e^3 b_{3u} * e^3 b_{2u} =$  plus combination:  $2.3 * 1.2 + 1.2 * 2.3$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ + \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI Vectors:

$$+1 \cdot |a2a0| \cdot |aa20| + 1 \cdot |0a2a| \cdot |aa20| - 1 \cdot |a2a0| \cdot |02aa| - 1 \cdot |0a2a| \cdot |02aa| \\ +1 \cdot |aa20| \cdot |a2a0| - 1 \cdot |02aa| \cdot |a2a0| + 1 \cdot |aa20| \cdot |0a2a| - 1 \cdot |02aa| \cdot |0a2a|$$

forms the following possible dimer combinations:

$$+\frac{1}{2} \cdot |a2a0aa20| + \frac{1}{2} \cdot |a220aaa0| + \frac{1}{2} \cdot |aaa0a220| + \frac{1}{2} \cdot |aa20a2a0| - \frac{1}{2} \cdot |0a2a02aa| - \frac{1}{2} \cdot |0aaa022a| \\ - \frac{1}{2} \cdot |022a0aaa| - \frac{1}{2} \cdot |02aa0a2a|$$

or, expressed in terms of symmetry:

$$= -4 \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \right| + 4 \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| + 4 \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| - 4 \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right| \\ - 4 \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} \right| + 4 \left| \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} \right| + 4 \left| \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} \right| - 4 \left| \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \right|$$

minus combination:  $e^3 b_{2u} * e^3 b_{3u} - e^3 b_{3u} * e^3 b_{2u} =$  minus combination:  $2.3 * 1.2 - 1.2 * 2.3$

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}$$

written in CI Vectors:

$$+1 \cdot |a2a0| \cdot |aa20| + 1 \cdot |0a2a| \cdot |aa20| - 1 \cdot |a2a0| \cdot |02aa| - 1 \cdot |0a2a| \cdot |02aa| \\ -1 \cdot |aa20| \cdot |a2a0| + 1 \cdot |02aa| \cdot |a2a0| - 1 \cdot |aa20| \cdot |0a2a| + 1 \cdot |02aa| \cdot |0a2a|$$

forms the following possible dimer combinations:

$$-\frac{1}{2} \cdot |0a2aaa20| + \frac{1}{2} \cdot |0a20aa2a| - \frac{1}{2} \cdot |aa2a0a20| + \frac{1}{2} \cdot |aa200a2a| + \frac{1}{2} \cdot |a2a002aa| - \frac{1}{2} \cdot |a2aa02a0| \\ + \frac{1}{2} \cdot |02a0a2aa| - \frac{1}{2} \cdot |02aaa2a0|$$

or, expressed in terms of symmetry:

$$\begin{aligned}
&= -4 \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \right| + 4 \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| + 4 \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| - 4 \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right| \\
&\quad - 4 \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} \right| + 4 \left| \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} \right| + 4 \left| \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} \right| - 4 \left| \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \right|
\end{aligned}$$


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plus combination:  $e^3 b_{2u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{2u} =$  plus combination:  $2.3 * 2.2 + 2.2 * 2.3$

$$\begin{aligned}
&+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\
&+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix}
\end{aligned}$$

written in CI Vectors:

$$\begin{aligned}
&+1 \cdot |a2a0| \cdot |aa20| + 1 \cdot |0a2a| \cdot |aa20| + 1 \cdot |a2a0| \cdot |02aa| + 1 \cdot |0a2a| \cdot |02aa| \\
&+1 \cdot |aa20| \cdot |a2a0| + 1 \cdot |02aa| \cdot |a2a0| + 1 \cdot |aa20| \cdot |0a2a| + 1 \cdot |02aa| \cdot |0a2a|
\end{aligned}$$

forms the following possible dimer combinations:

$$\begin{aligned}
&+ \frac{1}{2} \cdot |a2a0aa20| + \frac{1}{2} \cdot |a220aaa0| + \frac{1}{2} \cdot |aaa0a220| + \frac{1}{2} \cdot |aa20a2a0| + \frac{1}{2} \cdot |0a2a02aa| + \frac{1}{2} \cdot |0aaa022a| \\
&+ \frac{1}{2} \cdot |022a0aaa| + \frac{1}{2} \cdot |02aa0a2a|
\end{aligned}$$

or, expressed in terms of symmetry:

$$\begin{aligned}
&= -4 \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \right| + 4 \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| + 4 \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| - 4 \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right| \\
&\quad + 4 \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} \right| - 4 \left| \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} \right| - 4 \left| \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} \right| + 4 \left| \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \right|
\end{aligned}$$


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minus combination:  $e^3 b_{2u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{2u} =$  minus combination: 2.3 \* 2.2 - 2.2 \* 2.3

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix} \\ - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix}$$

written in CI Vectors:

$$+1 \cdot |a2a0| \cdot |aa20| + 1 \cdot |0a2a| \cdot |aa20| + 1 \cdot |a2a0| \cdot |02aa| + 1 \cdot |0a2a| \cdot |02aa| \\ -1 \cdot |aa20| \cdot |a2a0| - 1 \cdot |02aa| \cdot |a2a0| - 1 \cdot |aa20| \cdot |0a2a| - 1 \cdot |02aa| \cdot |0a2a|$$

forms the following possible dimer combinations:

$$-\frac{1}{2} \cdot |0a2aaa20| + \frac{1}{2} \cdot |0a20aa2a| - \frac{1}{2} \cdot |aa2a0a20| + \frac{1}{2} \cdot |aa200a2a| - \frac{1}{2} \cdot |a2a002aa| + \frac{1}{2} \cdot |a2aa02a0| \\ -\frac{1}{2} \cdot |02a0a2aa| + \frac{1}{2} \cdot |02aaa2a0|$$

or, expressed in terms of symmetry:

$$= -4 \left| \underbrace{a_g b_{3u} b_{2u} b_{1u}}_{a_u} \right| + 4 \left| \underbrace{a_g b_{1u} b_{2g} b_{3g}}_{a_u} \right| + 4 \left| \underbrace{a_g b_{3u} b_{1g} b_{2g}}_{a_u} \right| - 4 \left| \underbrace{b_{3u} b_{1u} b_{2g} a_u}_{a_u} \right| \\ + 4 \left| \underbrace{a_g b_{2u} b_{1g} b_{3g}}_{a_u} \right| - 4 \left| \underbrace{b_{2u} b_{1u} b_{3g} a_u}_{a_u} \right| - 4 \left| \underbrace{b_{3u} b_{2u} b_{1g} a_u}_{a_u} \right| + 4 \left| \underbrace{b_{1g} b_{2g} b_{3g} a_u}_{a_u} \right|$$

identical  $e^3 b_{3u} * e^3 b_{3u} =$  identical 1.2 \* 1.2

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \uparrow\uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \uparrow\uparrow & \uparrow & \uparrow & \uparrow\uparrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \uparrow\uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \uparrow\uparrow & \uparrow & \uparrow\uparrow & \uparrow\uparrow \end{pmatrix}$$

written in CI Vectors:

$$+1 \cdot |aa20| \cdot |aa20| - 1 \cdot |02aa| \cdot |aa20| - 1 \cdot |aa20| \cdot |02aa| + 1 \cdot |02aa| \cdot |02aa|$$

forms the following possible dimer combinations:

$$+1 \cdot |aa20aa20| + 1 \cdot |02aa02aa| - \frac{1}{8} \cdot |02aaaa20| + \frac{1}{8} \cdot |02a0aa2a| + \frac{1}{8} \cdot |022aaaa0| - \frac{1}{8} \cdot |0220aaaa| \\ + \frac{1}{8} \cdot |0aaaa220| - \frac{1}{8} \cdot |0aa0a22a| - \frac{1}{8} \cdot |0a2aa2a0| + \frac{1}{8} \cdot |0a20a2aa| + \frac{1}{8} \cdot |a2aa0a20| - \frac{1}{8} \cdot |a2a00a2a| \\ - \frac{1}{8} \cdot |a22a0aa0| + \frac{1}{8} \cdot |a2200aaa| - \frac{1}{8} \cdot |aaaa0220| + \frac{1}{8} \cdot |aaa0022a| + \frac{1}{8} \cdot |aa2a02a0| - \frac{1}{8} \cdot |aa2002aa|$$

or, expressed in terms of symmetry:

$$\begin{aligned}
&= -4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| - 2 \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| - 2 \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| - 2 \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\
&\quad + 2 \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| + 2 \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| - 2 \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| - 2 \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\
&\quad - 2 \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - 4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|
\end{aligned}$$


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plus combination:  $e^3 b_{3u} * i^3 b_{3u} + i^3 b_{3u} * e^3 b_{3u} =$  plus combination:  $1.2 * 2.2 + 2.2 * 1.2$

$$\begin{aligned}
&+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\
&+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix}
\end{aligned}$$

written in CI Vectors:

$$\begin{aligned}
&+1 \cdot |aa20| \cdot |aa20| - 1 \cdot |02aa| \cdot |aa20| + 1 \cdot |aa20| \cdot |02aa| - 1 \cdot |02aa| \cdot |02aa| \\
&+1 \cdot |aa20| \cdot |aa20| + 1 \cdot |02aa| \cdot |aa20| - 1 \cdot |aa20| \cdot |02aa| - 1 \cdot |02aa| \cdot |02aa|
\end{aligned}$$

forms the following possible dimer combinations:

$$+2 \cdot |aa20aa20| - 2 \cdot |02aa02aa|$$

or, expressed in terms of symmetry:

$$= -8 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| + 8 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|$$


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minus combination:  $e^3 b_{3u} * i^3 b_{3u} - i^3 b_{3u} * e^3 b_{3u} =$  minus combination: 1.2 \* 2.2 - 2.2 \* 1.2

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix} \\ - \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} - \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix}$$

written in CI Vectors:

$$+1 \cdot |aa20| \cdot |aa20| - 1 \cdot |02aa| \cdot |aa20| + 1 \cdot |aa20| \cdot |02aa| - 1 \cdot |02aa| \cdot |02aa| \\ -1 \cdot |aa20| \cdot |aa20| - 1 \cdot |02aa| \cdot |aa20| + 1 \cdot |aa20| \cdot |02aa| + 1 \cdot |02aa| \cdot |02aa|$$

forms the following possible dimer combinations:

or, expressed in terms of symmetry:

$$= -8 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| + 8 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|$$

identical  $i^3 b_{3u} * i^3 b_{3u} =$  identical 2.2 \* 2.2

$$+ \begin{pmatrix} \uparrow & - & \uparrow & - \\ \uparrow & \downarrow\downarrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & \uparrow & - \\ \downarrow\downarrow & \uparrow & \uparrow & \downarrow\downarrow \end{pmatrix} + \begin{pmatrix} \uparrow & - & - & \uparrow \\ \uparrow & \downarrow\downarrow & \downarrow\downarrow & \uparrow \end{pmatrix} + \begin{pmatrix} - & \uparrow & - & \uparrow \\ \downarrow\downarrow & \uparrow & \downarrow\downarrow & \uparrow \end{pmatrix}$$

written in CI Vectors:

$$+1 \cdot |aa20| \cdot |aa20| + 1 \cdot |02aa| \cdot |aa20| + 1 \cdot |aa20| \cdot |02aa| + 1 \cdot |02aa| \cdot |02aa|$$

forms the following possible dimer combinations:

$$+1 \cdot |aa20aa20| + 1 \cdot |02aa02aa| + \frac{1}{8} \cdot |02aaaa20| - \frac{1}{8} \cdot |02a0aa2a| - \frac{1}{8} \cdot |022aaaa0| + \frac{1}{8} \cdot |0220aaaa| \\ - \frac{1}{8} \cdot |0aaaa220| + \frac{1}{8} \cdot |0aa0a22a| + \frac{1}{8} \cdot |0a2aa2a0| - \frac{1}{8} \cdot |0a20a2aa| - \frac{1}{8} \cdot |a2aa0a20| + \frac{1}{8} \cdot |a2a00a2a| \\ + \frac{1}{8} \cdot |a22a0aa0| - \frac{1}{8} \cdot |a2200aaa| + \frac{1}{8} \cdot |aaaa0220| - \frac{1}{8} \cdot |aaa0022a| - \frac{1}{8} \cdot |aa2a02a0| + \frac{1}{8} \cdot |aa2002aa|$$

or, expressed in terms of symmetry:

$$= -4 \left| \underbrace{a_g b_{3u} b_{1u} b_{2g}}_{a_g} \right| + 2 \left| \underbrace{a_g b_{3u} b_{2u} b_{1g}}_{a_g} \right| + 2 \left| \underbrace{a_g b_{3u} b_{3g} a_u}_{a_g} \right| + 2 \left| \underbrace{a_g b_{2u} b_{2g} a_u}_{a_g} \right| \\ - 2 \left| \underbrace{a_g b_{1g} b_{2g} b_{3g}}_{a_g} \right| - 2 \left| \underbrace{b_{3u} b_{2u} b_{1u} a_u}_{a_g} \right| + 2 \left| \underbrace{b_{3u} b_{1g} b_{1u} b_{3g}}_{a_g} \right| + 2 \left| \underbrace{b_{2u} b_{1g} b_{1u} b_{2g}}_{a_g} \right| \\ + 2 \left| \underbrace{b_{1u} b_{2g} b_{3g} a_u}_{a_g} \right| - 4 \left| \underbrace{b_{2u} b_{1g} b_{3g} a_u}_{a_g} \right|$$

